

## Day 33: Apply Discount

### Day 33 — Apply Discount

#### 1. Day Number

Day 33

#### 2. Topic Name

Percentage and if-else

Today you will learn how to calculate a percentage and use a condition to decide whether a discount should be applied.

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#### 3. Connection

Yesterday, you calculated the **total price of the cart**.

Today, you will take that cart total and apply a simple discount:

```
Products
  ↓
Cart
  ↓
Cart Total
  ↓
Check Discount Condition
  ↓
Final Amount
```

So your shopping-cart program is becoming more realistic.

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#### 4. Important Topics

Today's important concepts are:

- Percentage
- Arithmetic
- if-else
- Comparison operators
- Functions
- Returning a calculated value

The main idea is:

```
If condition is true
  do something
otherwise
  do something else
```

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## 5. Foundational Notes

### A. What is a percentage?

A percentage means **a part out of 100**.

For example:

10% means 10 out of 100.

So:

10% of 200 = 20

Conceptually:

percentage amount = original amount  $\times$  percentage / 100

For example:

20% of 100  
=  $100 \times 20 / 100$   
= 20

---

### B. What is a discount?

A discount reduces the original price.

Suppose:

Original price = 1000  
Discount = 10%

First calculate the discount amount:

10% of 1000 = 100

Then subtract it:

1000 - 100 = 900

So:

Final price = 900

Think of it as:

Original Total  
|  
v  
Calculate Discount  
|  
v

Subtract Discount

|

v

Final Total

## C. What does if-else do?

if-else lets your program make a decision.

Basic structure:

```
if condition:
    # run when condition is True
else:
    # run when condition is False
```

For example:

```
age = 20

if age >= 18:
    print("Adult")
else:
    print("Minor")
```

The program checks:

Is age >= 18?

If yes:

Adult

Otherwise:

Minor

## D. Comparison operators

You already need to recognize these:

| Operator | Meaning                  |
|----------|--------------------------|
| >        | Greater than             |
| <        | Less than                |
| >=       | Greater than or equal to |
| <=       | Less than or equal to    |
| ==       | Equal to                 |

| Operator | Meaning      |
|----------|--------------|
| $\neq$   | Not equal to |

Today's condition specifically says:

| discount only when total is **greater than 500**

So pay close attention to the difference between:

$> 500$

and

$\geq 500$

They are **not** the same.

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## 6. Easy Example

Let's use a different example from today's problem.

Suppose a shop gives a **20% discount when the amount is greater than 1000**.

Imagine:

amount = 1500

Check:

$1500 > 1000$

This is:

True

Calculate:

20% of 1500 = 300

Then:

$1500 - 300 = 1200$

So the discounted amount would be:

1200

But if:

amount = 800

the condition:

`800 > 1000`

is false.

Therefore:

`Final amount = 800`

No discount is applied.

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## 7. Problem Statement

Create a Python program that:

| Asks the user for the cart total.

Then:

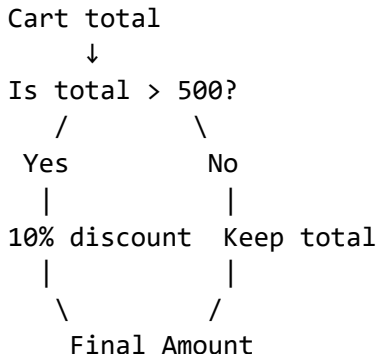
| If the cart total is greater than 500, apply a 10% discount.

Otherwise:

| Keep the original cart total.

Finally, display the final amount.

### Example logic



## 8. Concepts Used

You will use:

### Variables

To store values such as:

`cart total`  
`discount`  
`final total`

### Arithmetic

You need:

```
*  
/  
-
```

for percentage calculation and subtraction.

## Comparison

You need to check whether:

```
cart total > 500
```

## if-else

Used to choose between:

```
apply discount
```

and:

```
keep original amount
```

## Function

Instead of putting all logic directly in the program, place the discount calculation inside a function.

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## 9. Thought Process

Before writing Python, think about the problem in small steps.

### Step 1: Get the cart total

You need a number representing the total amount.

For example:

```
800
```

### Step 2: Check the condition

Ask:

```
Is 800 greater than 500?
```

Yes.

Therefore, a discount should be applied.

### Step 3: Calculate 10%

Think:

```
discount = 10% of cart total
```

For 800:

10% of 800 = 80

#### Step 4: Calculate the final amount

800 - 80 = 720

#### Step 5: Return/display the result

The program should produce:

Final total: 720

The overall reasoning is:

```
Input total
↓
Check total > 500
↓
+-----+-----+
|               |               |
| True           | False        |
|               |               |
| Calculate 10%   | No discount  |
|               |               |
| Subtract it     | Keep total   |
|               |               |
+-----+-----+
      |
      v
    Final total
```

## 10. Pseudocode

Write the logic in plain language before Python:

```
START

DEFINE a function that receives cart total

    IF cart total is greater than 500
        calculate 10 percent discount
        subtract discount from cart total
        return discounted total
    ELSE
        return original cart total

ASK user for cart total

CONVERT input into a number
```

```
CALL the discount function

STORE returned value

PRINT final total

END
```

Notice that this explains the algorithm without giving you the full Python implementation.

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## 11. Suggested Solving Approach — Functional Approach

Use a **function** to handle the discount calculation.

Conceptually:

```
main program
  |
  | cart_total
  v
discount function
  |
  | checks > 500
  |
  | calculates result
  v
return final_total
  |
  v
main program prints result
```

A good function design would look conceptually like:

```
calculate_discount(total)

Input:
  total

Process:
  check discount condition

Output:
  final total
```

The function should have **one simple responsibility**:

| Take a total and determine the final price after the discount rule.

This is better than mixing input, calculation, and printing into one large block of logic.

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## 12. Easy Edge Cases

### Edge Case 1: Total exactly 500

Suppose:

total = 500

The rule says:

greater than 500

Check:

$500 > 500$

This is:

False

Therefore:

No discount

Final amount:

500

This is an important boundary condition.

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### Edge Case 2: Total below 500

Example:

total = 400

Check:

$400 > 500$

False.

So:

Final amount = 400

---

### Edge Case 3: Total 0

Example:

total = 0

Check:

```
0 > 500
```

False.

Therefore:

```
Final amount = 0
```

Your program should still behave correctly.

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## 13. Expected Input

One cart-total value.

For example:

```
Enter cart total: 800
```

For today's beginner exercise, you can assume the user enters a valid number.

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## 14. Expected Output

### Example 1 — Discount applied

Input:

```
Enter cart total: 800
```

Calculation:

```
10% of 800 = 80  
800 - 80 = 720
```

Expected output:

```
Final total: 720
```

### Example 2 — No discount

Input:

```
Enter cart total: 450
```

Expected output:

```
Final total: 450
```

### Example 3 — Exactly 500

Input:

Enter cart total: 500

Expected output:

Final total: 500

## 15. Hint Only

Think about these three questions while coding:

1. Which comparison operator represents **greater than 500**?
2. How can you calculate **10% of total** using multiplication and division?
3. What should your function return when the condition is false?

A useful mental model is:

```
discount_amount = percentage of total  
  
final_total = total - discount_amount
```

Then place that calculation inside the correct branch of your if-else.

**Do not forget:** 500 itself should **not** receive the discount because the requirement says **greater than 500**, not greater than or equal to 500.